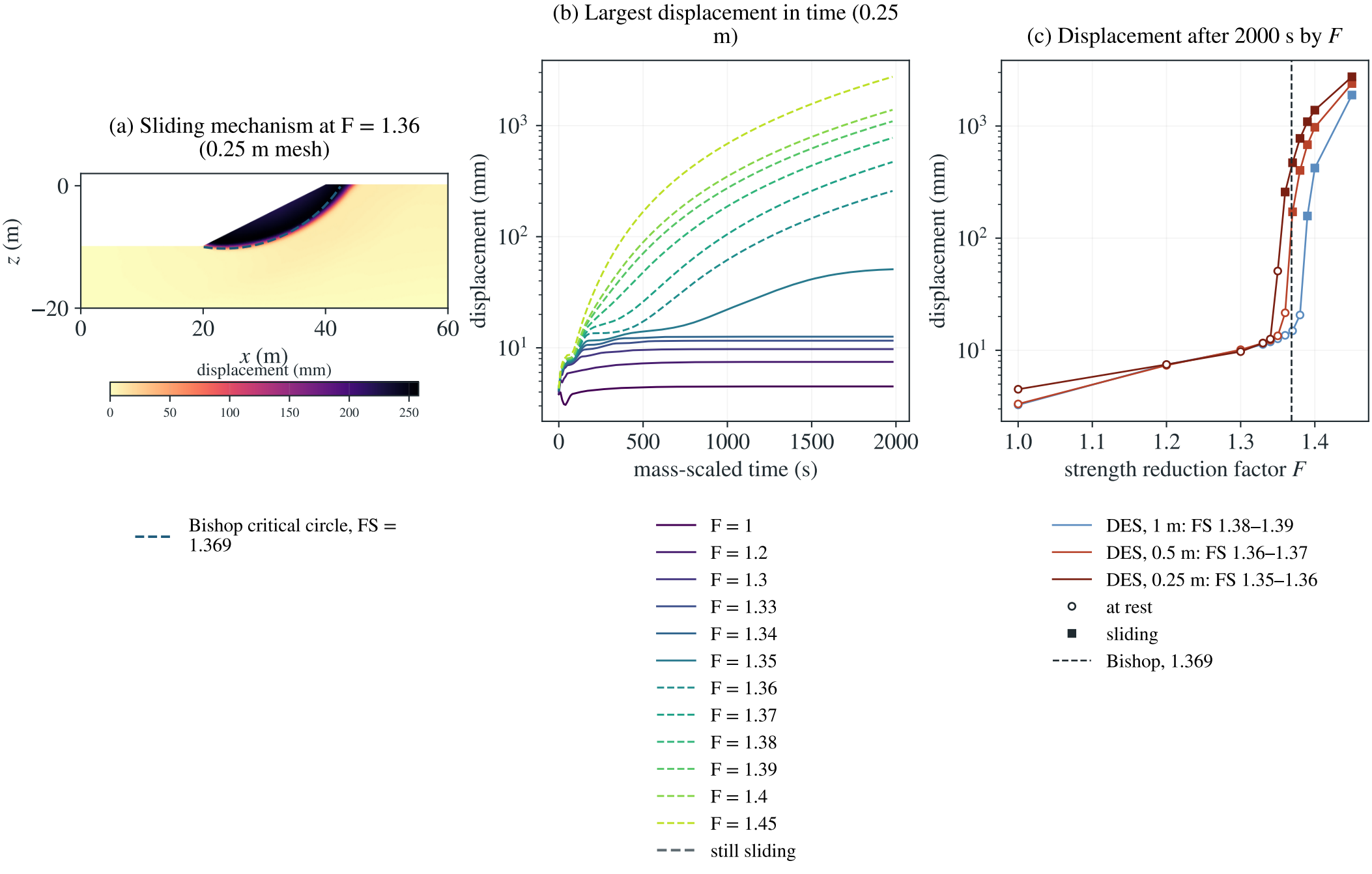

(a) Sliding mechanism at F = 1.36
(0.25 m mesh)
z (m)
0
−20
0
20
40
60
x (m)
displacement (mm)
0
50
100
150
200
250
Bishop critical circle, FS = 1.369
(b) Largest displacement in time (0.25 m)
displacement (mm)
10^3
10^2
10^1
0
500
1000
1500
2000
mass-scaled time (s)
F = 1
F = 1.2
F = 1.3
F = 1.33
F = 1.34
F = 1.35
F = 1.36
F = 1.37
F = 1.38
F = 1.39
F = 1.4
F = 1.45
still sliding
(c) Displacement after 2000 s by F
displacement (mm)
10^3
10^2
10^1
1.0
1.1
1.2
1.3
1.4
strength reduction factor F
DES, 1 m: FS 1.38–1.39
DES, 0.5 m: FS 1.36–1.37
DES, 0.25 m: FS 1.35–1.36
at rest
sliding
Bishop, 1.369